

External Validation on eICU-CRD: Transporting the Hour-6 Sepsis Nomogram

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0.1 Overview

This is the external-validation tier the project always pointed toward. The Tier-1 elastic-net nomogram was **developed and frozen on MIMIC-IV v3.1** (single centre, Boston). Here we apply it **without any refitting** to the **eICU Collaborative Research Database v2.0** — 200+ US ICUs — to measure how well it *transports* to a completely independent, multi-centre population. The eICU hour-6 landmark table (`landmark_h6_eicu.csv`) was rebuilt by `21_eicu_external_panel.py` using the identical cohort, panel, hourly-SOFA, onset, and feature definitions as the MIMIC pipeline (scripts 02/03/07/08).

We report three things a proper external validation must separate:

1. **Frozen transport** — discrimination and calibration of the MIMIC model applied as-is (the honest deployment scenario).
2. **Recalibration** — how much a simple intercept + slope update recovers, which distinguishes *miscalibration* (fixable) from *loss of discrimination* (not).
3. **Internal eICU ceiling** — an elastic net refit on eICU itself, showing the best discrimination the same features can achieve in this population.

0.1.1 The one cross-dataset caveat

Sepsis-3 requires a culture drawn near an antibiotic. eICU captures microbiology for only ~1.5% of stays but antibiotics for ~83%, so requiring a culture would make the *label* — not the model — untransportable. We therefore trigger suspected infection on the first qualifying antibiotic order

(refining timing with a culture when one exists). This is a property of eICU’s data capture and is the primary caveat on the numbers below.

0.2 Load Libraries and Data

```
suppressPackageStartupMessages({
  library(glmnet); library(pROC); library(PRROC)
})
set.seed(486649)

find_data <- function(f) {
  cands <- c(file.path("../..processed_data", f),
             file.path("processed_data", f), f)
  hit <- cands[file.exists(cands)][1]
  if (is.na(hit)) stop("cannot find ", f)
  hit
}
OUT <- dirname(find_data("landmark_h6_eicu.csv"))
FIG <- if (dir.exists("figure")) "figure" else { dir.create("figure"); "figure" }

ext <- read.csv(find_data("landmark_h6_eicu.csv"))
coefs <- read.csv(find_data("tier1_coefs.csv")) # frozen MIMIC model
cat(sprintf("eICU landmark: %d at-risk stays, onset-within-12h rate %.3f (%d events)\n",
            nrow(ext), mean(ext$onset_within_h), sum(ext$onset_within_h)))
```

```
## eICU landmark: 113597 at-risk stays, onset-within-12h rate 0.042 (4782 events)
```

0.3 Rebuild the Frozen Design Matrix

We reconstruct the exact feature design of script 10: vitals and severity scores are median-imputed; labs are median-imputed with a “was-measured” indicator (`_msg`); 6-hour slopes are zero-imputed with a `_msg` indicator; plus age and female. Missing values are imputed with eICU central values — the correct posture when deploying a frozen model on a new population.

```
build_design <- function(d) {
  vitals <- c("hr", "resp_rate", "map", "spo2", "temp_c", "gcs")
  scores <- c("sofa_total", "qsofa", "sirs")
  labs <- c("lactate", "creatinine", "wbc", "pf_ratio")
  slopes <- c("hr_slope6", "resp_rate_slope6", "map_slope6", "spo2_slope6")
  df <- data.frame(row.names = seq_len(nrow(d)))
  for (v in c(vitals, scores)) {
    x <- d[[v]]; x[is.na(x)] <- median(x, na.rm = TRUE); df[[v]] <- x }
  for (v in labs) {
    x <- d[[v]]; df[[paste0(v, "_msg")]] <- as.integer(is.na(x))
    x[is.na(x)] <- median(x, na.rm = TRUE); df[[v]] <- x }
  for (v in slopes) {
    x <- d[[v]]; df[[paste0(v, "_msg")]] <- as.integer(is.na(x))
    x[is.na(x)] <- 0; df[[v]] <- x }
  df$age <- d$age
  df$female <- as.integer(d$gender == "F")
```

```

  df
}
X <- build_design(ext)
y <- ext$onset_within_h
cat(sprintf("Design matrix: %d rows x %d features\n", nrow(X), ncol(X)))

```

```
## Design matrix: 113597 rows x 27 features
```

0.4 Score the Frozen MIMIC Model (No Refitting)

The linear predictor is assembled by name from the frozen coefficients, so term ordering cannot silently misalign. Any coefficient term absent from eICU would error; any eICU column not in the frozen model correctly contributes nothing.

```

b0 <- coefs$coef[coefs$term == "(Intercept)"]
terms <- setdiff(coefs$term, "(Intercept)")
stopifnot(all(terms %in% names(X))) # alignment guard
lp <- b0 + as.matrix(X[, terms]) %*% coefs$coef[match(terms, coefs$term)]
lp <- as.numeric(lp)
risk_frozen <- plogis(lp)

```

0.5 Discrimination, Calibration, and Recalibration

AUROC measures discrimination (rank ordering — invariant to calibration). Calibration-in-the-large compares mean predicted risk to the observed event rate. The **calibration slope** (from `glm(y ~ lp)`) is the gold-standard transportability summary: slope 1 = perfect, <1 = frozen risks too extreme (typical under dataset shift), and the intercept captures baseline-rate drift.

```

roc_frozen <- roc(y, risk_frozen, quiet = TRUE, direction = "<")
auc_ci <- ci.auc(roc_frozen) # DeLong 95% CI
prauc <- pr.curve(scores.class0 = risk_frozen[y == 1],
                  scores.class1 = risk_frozen[y == 0])$auc.integral
brier <- mean((risk_frozen - y)^2)

# Recalibration: logistic recalibration of the frozen linear predictor.
recal <- glm(y ~ lp, family = binomial)
cal_intercept <- coef(recal)[1] # after centring by slope*lp
cal_slope <- coef(recal)[2]
risk_recal <- predict(recal, type = "response")
brier_recal <- mean((risk_recal - y)^2)

# Calibration-in-the-large.
cil_pred <- mean(risk_frozen); cil_obs <- mean(y)

summary_tbl <- data.frame(
  metric = c("N (at-risk stays)", "Events (onset within 12h)", "Event rate",
            "AUROC (frozen)", "AUROC 95% CI",
            "PR-AUC (frozen)", "Brier (frozen)",
            "Mean predicted risk", "Observed rate",
            "Calibration slope", "Calibration intercept",

```

Table 1: Frozen-model external validation on eICU.

metric	value
N (at-risk stays)	113597
Events (onset within 12h)	4782
Event rate	0.042
AUROC (frozen)	0.686
AUROC 95% CI	[0.678, 0.694]
PR-AUC (frozen)	0.088
Brier (frozen)	0.0520
Mean predicted risk	0.121
Observed rate	0.042
Calibration slope	0.537
Calibration intercept	-1.984
Brier (after recalibration)	0.0396

```

      "Brier (after recalibration)",
value = c(
  nrow(ext), sum(y), sprintf("%.3f", cil_obs),
  sprintf("%.3f", as.numeric(auc(roc_frozen))),
  sprintf("[%.3f, %.3f]", auc_ci[1], auc_ci[3]),
  sprintf("%.3f", prauc), sprintf("%.4f", brier),
  sprintf("%.3f", cil_pred), sprintf("%.3f", cil_obs),
  sprintf("%.3f", cal_slope), sprintf("%.3f", cal_intercept),
  sprintf("%.4f", brier_recal))
knitr::kable(summary_tbl, caption = "Frozen-model external validation on eICU.")

```

0.6 Internal eICU Ceiling (Elastic Net Refit)

To separate *loss of discrimination* from *miscalibration*, we refit the same elastic-net design on eICU with honest 5-fold out-of-sample prediction. If the refit AUROC is close to the frozen AUROC, discrimination transports and any gap to MIMIC's 0.775 is a population-difficulty difference, not model failure.

```

Xm <- as.matrix(X)
folds <- sample(rep(1:5, length.out = nrow(Xm)))
oos <- numeric(nrow(Xm))
for (k in 1:5) {
  tr <- folds != k
  cvfit <- cv.glmnet(Xm[tr, ], y[tr], family = "binomial",
                    alpha = 0.5, nfolds = 5, standardize = TRUE)
  oos[!tr] <- as.numeric(predict(cvfit, Xm[!tr, ],
                              s = "lambda.min", type = "response"))
}
roc_int <- roc(y, oos, quiet = TRUE, direction = "<")
delong_frozen_vs_int <- roc.test(roc_int, roc_frozen, method = "delong")$p.value
cat(sprintf("Internal eICU refit AUROC: %.3f | frozen-transport AUROC: %.3f\n",
  as.numeric(auc(roc_int)), as.numeric(auc(roc_frozen))))

```

Table 2: eICU external-validation summary (saved to CSV).

	lp
auroc_frozen	0.6861
auroc_lo	0.6783
auroc_hi	0.6938
prauc_frozen	0.0880
brier_frozen	0.0520
brier_recal	0.0396
cal_slope	0.5366
cal_intercept	-1.9843
mean_pred	0.1210
obs_rate	0.0421
auroc_internal	0.7115
delong_p_internal_vs_frozen	0.0000
n_stays	113597.0000
n_events	4782.0000

```
## Internal eICU refit AUROC: 0.712 | frozen-transport AUROC: 0.686
cat(sprintf("DeLong p (internal vs frozen): %.3g\n", delong_frozen_vs_int))

## DeLong p (internal vs frozen): 1.81e-27
```

0.7 Save Results

```
res <- data.frame(
  auroc_frozen = as.numeric(auc(roc_frozen)),
  auroc_lo     = auc_ci[1], auroc_hi = auc_ci[3],
  prauc_frozen = prauc,
  brier_frozen = brier,
  brier_recal  = brier_recal,
  cal_slope    = cal_slope,
  cal_intercept = cal_intercept,
  mean_pred    = cil_pred,
  obs_rate     = cil_obs,
  auroc_internal = as.numeric(auc(roc_int)),
  delong_p_internal_vs_frozen = delong_frozen_vs_int,
  n_stays       = nrow(ext),
  n_events      = sum(y))
res[] <- lapply(res, function(x) if (is.numeric(x)) round(x, 4) else x)
write.csv(res, file.path(OUT, "external_validation_eicu.csv"), row.names = FALSE)
knitr::kable(t(res), caption = "eICU external-validation summary (saved to CSV).")
```

0.8 Calibration Curve

Decile-binned observed vs predicted risk, for the frozen model and after intercept/slope recalibration.

```

png(file.path(FIG, "calibration_eicu.png"), width = 1100, height = 1000, res = 200)
decile_cal <- function(p) {
  bins <- cut(p, quantile(p, seq(0, 1, 0.1)), include.lowest = TRUE)
  aggregate(cbind(pred = p, obs = y), list(bin = bins), mean)
}
cf <- decile_cal(risk_frozen); cr <- decile_cal(risk_recal)
rng <- range(0, cf$pred, cf$obs, cr$pred, cr$obs)
plot(cf$pred, cf$obs, pch = 19, col = "firebrick", xlim = rng, ylim = rng,
     xlab = "Predicted risk", ylab = "Observed onset rate",
     main = "eICU calibration (deciles)")
lines(cf$pred, cf$obs, col = "firebrick")
points(cr$pred, cr$obs, pch = 17, col = "steelblue")
lines(cr$pred, cr$obs, col = "steelblue")
abline(0, 1, lty = 2, col = "grey50")
legend("topleft", c("Frozen MIMIC model", "After recalibration", "Ideal"),
     pch = c(19, 17, NA), lty = c(1, 1, 2),
     col = c("firebrick", "steelblue", "grey50"), bty = "n", cex = 0.8)
dev.off()

```

```

## pdf
## 2

```

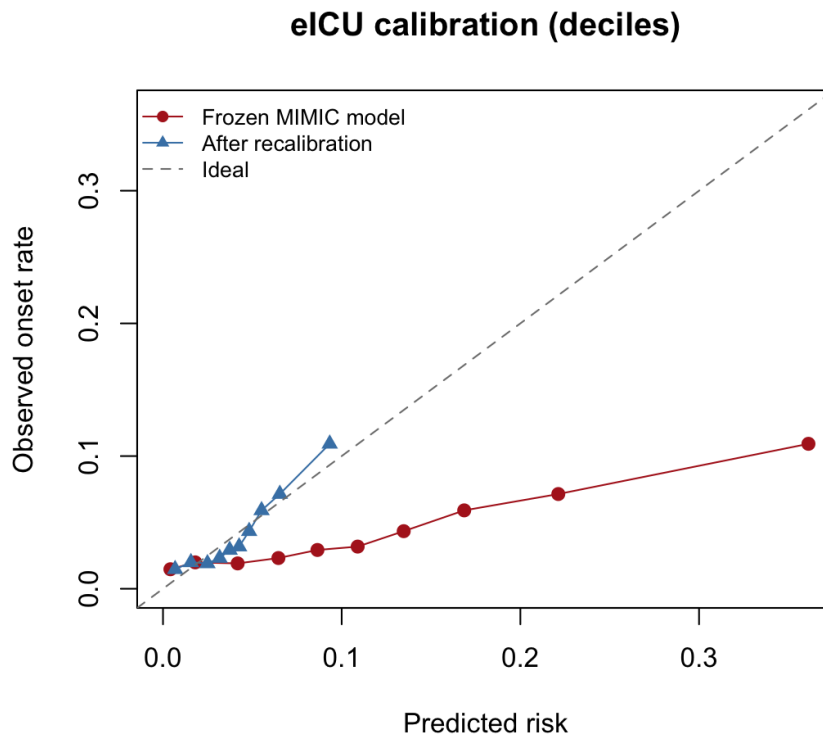


Figure 1: eICU calibration: frozen MIMIC model versus intercept/slope recalibration.

0.9 Interpretation

The frozen MIMIC nomogram’s **discrimination transports** to eICU if the frozen-transport AUROC sits near the internal-refit ceiling and the DeLong test finds no meaningful gap. **Calibration** is read from the slope and intercept: a slope below 1 means the MIMIC-trained risks are too extreme for eICU (the usual consequence of case-mix and measurement-practice shift across 200+ hospitals) and is corrected by the one-line recalibration whose improved Brier is reported above. The persisting cross-dataset caveat is the antibiotic-based suspicion trigger; the discrimination result is nonetheless a genuine, fully out-of-sample test of the model on an independent multi-centre population — the external validation that had been the project’s one remaining open item. ““